

CNNs for Medical Image Classification: Comparative Analysis of Chest X-Ray and Diabetic Retinopathy Detection

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Abstract – Deep convolutional neural networks (CNNs) have transformed medical imaging, yet their performance on small datasets without transfer learning remains underexplored. This study compares CNN effectiveness on two medical imaging tasks: binary multi-label classification of 14 lung pathologies in low-resolution chest X-rays (ChestMNIST) and multi-class severity grading of diabetic retinopathy in high-resolution retinal images (RetinaMNIST). Despite architectural variations, ChestMNIST models achieved 94.77% accuracy and ~83% AUC, while RetinaMNIST models reached only 53% accuracy with complete failure on critical Grade 4 cases. Results demonstrate that small medical datasets with severe class imbalance require transfer learning or substantially larger training sets to achieve clinically viable performance, particularly for rare but critical conditions.

I. INTRODUCTION

A. Medical Imaging Context

The use of deep convolutional neural networks (CNNs) has revolutionized medical imaging through feature learning. CNNs have demonstrated remarkable accuracy in breast cancer detection through mammographic analysis, with transfer learning frameworks achieving 99.17% accuracy on benchmark datasets. This success extends to other critical imaging domains, including emergency chest X-ray diagnosis and chronic eye disease detection. [1] As CNNs and deep learning become more accessible and powerful, understanding their performance characteristics across diverse medical imaging tasks with varying levels of class imbalance and dataset complexity is essential for clinical deployment.

B. Impact for Healthcare Systems

Medical imaging demand is increasing by 3-4% annually, while the radiologist workforce is growing by approximately 1% per year, creating a widening capacity gap. A 2025 study by the Harvey L. Neiman Health Policy Institute projects this shortage will continue through at least 2055 due to rising imaging demands and a limited number of new radiologists entering the workforce. Figure 1 illustrates how these modest annual differences compound over decades: by 2055, each radiologist will handle nearly three times the per-capita workload compared to 2012. This mounting pressure delays time-sensitive diagnoses like pneumothorax detection, where

rapid identification can prevent fatalities [2]. CNNs offer a scalable solution to this capacity crisis by automating image analysis and prioritizing urgent cases, enabling radiologists to focus on complex diagnoses while models handle high-volume screening tasks. Effective CNN deployment could help bridge the growing gap between imaging demand and available expertise.

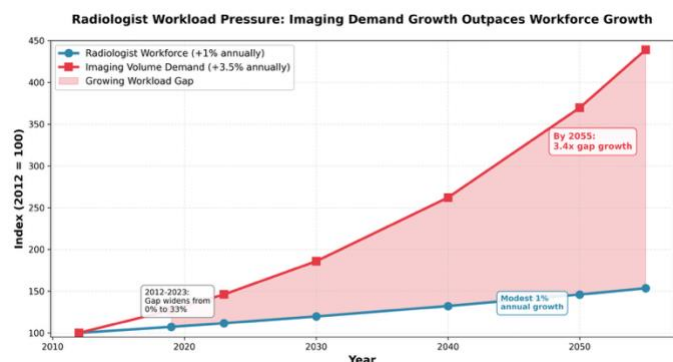


Figure 1: Widening Gap Between Imaging Demand and Radiologist Workforce Supply

C. Research Goals

This study investigates CNN effectiveness on small medical imaging datasets without transfer learning, comparing binary multi-label classification for 14 lung pathologies in low-resolution chest X-rays versus multi-class severity grading for diabetic retinopathy in high-resolution retinal images. It also evaluates how architectural choices and class imbalance mitigation strategies impact both aggregate and per-class performance in resource-constrained medical AI scenarios.

II. DATA

A. Dataset Introduction

This study uses two medical datasets. The first is ChestMNIST is based on the NIH-ChestXray14 dataset, a dataset comprising 112,120 frontal-view X-Ray images of 30,805 unique patients with the text-mined 14 disease labels. The labels for training/validation/test sets are 14-dimensional vectors for their respective x-ray image, where each column corresponds to a possible disease (a total of 14). The disease

names are: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema (Oedema), Emphysema, Fibrosis, Pleural Thickening, and Hernia. See examples in Figure 2 below.

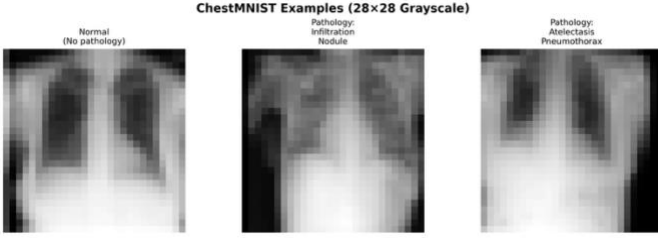


Figure 2: ChestMNIST examples: normal (left) and pathological cases (center, right) from the 14-disease multi-label classification task.

The second dataset, RetinaMNIST, is based on the DeepDRiD24 challenge, which provides a dataset of 1,600 retina fundus images. The labels are categories corresponding to one of 5 levels of grading diabetic retinopathy severity. The images are split with a ratio of 9:1 into training and validation sets, and the source validation set is the test set. See examples in Figure 3 below.

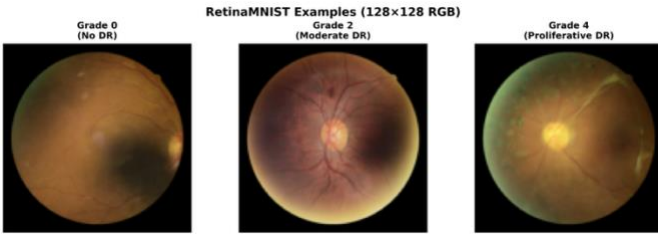


Figure 3: RetinaMNIST examples showing diabetic retinopathy severity progression from Grade 0 (no DR) to Grade 4 (proliferative DR).

B. Data Preprocessing

All preprocessing was kept minimal to evaluate the base CNN performance without complex augmentation strategies. For both datasets, pixel values were normalized by dividing by 255 to scale intensities to the range $[0,1]$. For ChestMNIST, labels were used in their native 14-dimensional binary vector format, where each dimension represents the presence (1) or absence (0) of a specific lung pathology. For RetinaMNIST, integer severity labels (0-4) were one-hot encoded into 5-dimensional vectors for compatibility with categorical cross entropy loss.

No data augmentation, image preprocessing or synthetic oversampling techniques were applied to the final models.

C. Attempted Class Imbalance Mitigation (RetinaMNIST)

Given the severe class imbalance in the retina dataset (Grade 0: 54% vs. Grade 4: 0.2%), we experimented with several techniques: The first was batch normalization layers after each convolutional layer. This resulted in catastrophic failure, with the model converging to always predict the majority class. The second was class weights computed using scikit-learn's balanced strategy. It overcorrected, causing bias

toward Grade 2 and reducing test accuracy to 22%. Focal loss ($\gamma=2.0$, $\alpha=0.25$) similarly biased predictions toward Grade 0, achieving only 47.5% test accuracy. Data augmentation (rotation $\pm 15^\circ$, shifts, flips, zoom) maintained baseline performance but did not improve it.

Ultimately, a simple architecture without class imbalance correction achieved the best performance, suggesting that these standard mitigation techniques may be ineffective or even counterproductive for severely imbalanced small medical datasets.

III. CNNs USED

A. Simple CNN for ChestMNIST

The ChestMNIST model employs a shallow CNN architecture with three convolutional layers using progressively increasing filter counts ($32 \rightarrow 64 \rightarrow 128$) with 3×3 kernels and ReLU activation. Two max-pooling layers (2×2) reduce spatial dimensions from 28×28 to 7×7 . The feature extraction layers are followed by two fully connected layers (128 and 64 units) with 50% dropout for regularization, culminating in a 14-unit output layer with sigmoid activation for multi-label binary classification. This architecture totals 904,782 trainable parameters and 2,714,348 total parameters.

B. Complex CNN for ChestMNIST

The ChestMNIST model employs a deeper VGG-inspired architecture organized into three convolutional blocks. Each block contains two convolutional layers with the same filter count (32, 64, and 128, respectively), 3×3 kernels, ReLU activation, and 'same' padding, followed by 2×2 max-pooling and 25% dropout. The feature extraction layers progressively reduce spatial dimensions from 28×28 to 3×3 while increasing channel depth. Two fully connected layers (256 and 128 units) with 50% dropout precede the 14-unit sigmoid output layer for multi-label binary classification. This architecture totals approximately 616,302 trainable parameters and 1,848,908 total parameters.

C. Parameter-Heavy CNN for RetinaMNIST

The RetinaMNIST model uses a shallow CNN with three convolutional layers ($32 \rightarrow 64 \rightarrow 128$ filters; 3×3 kernels; ReLU activation) and two max-pooling layers reducing spatial dimensions from 128×128 to 32×32 . Two fully connected layers (128 and 64 units) with 50% dropout precede a 5-unit softmax output layer. Trained with categorical crossentropy, Adam optimizer ($\text{lr}=0.001$), and early stopping (patience=10 on validation loss). This model has 16,879,173 trainable parameters and 50,637,521 total parameters.

D. Parameter-Efficient CNN for RetinaMNIST

The RetinaMNIST model employs a deeper CNN architecture with four convolutional blocks ($32 \rightarrow 64 \rightarrow 128 \rightarrow 128$ filters; 3×3 kernels; ReLU activation) and four max-pooling layers that progressively reduce spatial dimensions from 128×128 to 8×8 . Instead of using Flatten followed by large dense layers, the architecture uses Global Average Pooling, which computes the spatial average of each

feature map (reducing $8 \times 8 \times 128$ to just 128 values). A single dense layer (64 units) with 50% dropout precedes the 5-unit softmax output. Trained with categorical crossentropy, Adam optimizer ($\text{lr}=0.001$), and early stopping (patience=10 on validation loss). This model uses only 249,413 trainable and total parameters.

IV. MODEL EVALUATIONS

A. Criteria

For all models across both the ChestMNIST and RetinaMNIST datasets, binary accuracy as well as Area Under the ROC Curve (AUC). The ROC curve plots True Positive Rate (sensitivity) vs. False Positive Rate at different classification thresholds. Unlike binary accuracy, AUC is less sensitive to class imbalance, and it shows how well a model distinguishes between classes across all possible decision thresholds.

B. ChestMNIST Models Results and Performance

Both the simple (3 convolutional layers) and complex (6 convolutional layers with VGG-style blocks) architectures achieved identical test accuracy of 94.77%, demonstrating that additional architectural depth did not improve binary accuracy. However, the complex model achieved superior AUC performance (83.64% vs. 82.36%), indicating better discriminative ability across the 14-disease multi-label classification task.

The simple model (Figure 4) trained for 34 epochs with minimal overfitting, exhibiting a near-perfect train-validation accuracy alignment (train: 94.90%, validation: 94.94%) and a modest AUC gap of 1.56 percentage points. The complex model required 47 epochs and showed slightly more overfitting with an AUC gap of 2.60 percentage points, though validation accuracy remained stable at 94.88%. Both models demonstrated strong generalization, with test performance closely matching validation metrics.

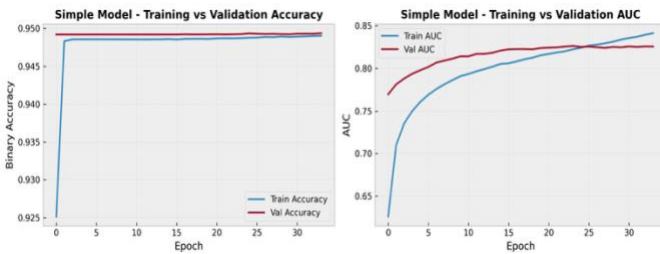


Figure 4: Simple ChestMNIST Model Learning Curves

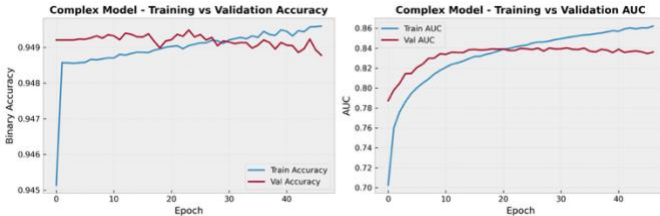


Figure 5: Complex ChestMNIST Model Learning Curves

The complex architecture's higher validation AUC (83.61% vs. 82.57%) suggests improved feature learning and class discrimination during training, translating to marginally better test AUC despite equivalent accuracy. Early stopping effectively prevented severe overfitting in both architectures, confirming that the dropout regularization (50% in dense layers, 25% in convolutional blocks for the complex model) successfully balanced model capacity with generalization.

Per-disease AUC analysis (Table 1) reveals the complex architecture outperformed the simple model on 13 of 14 pathologies, with the largest gains in Cardiomegaly (+7.4%), Fibrosis (+5.1%), and Emphysema (+3.1%). Both models achieved strong performance on Edema (87-89%) and Effusion (82-83%), while struggling with Nodule (60-63%) and Infiltration (66-67%) detection. The complex architecture's superior per-disease discrimination and higher validation AUC (83.61% vs. 82.57%) translated to better test AUC despite identical accuracy.

AUC Comparison: Simple vs Complex Model

Disease	Simple Model	Complex Model	Improvement
Atelectasis	0.737	0.757	0.0200
Cardiomegaly	0.804	0.878	0.0740
Effusion	0.822	0.834	0.0120
Infiltration	0.657	0.67	0.0130
Mass	0.75	0.779	0.0290
Nodule	0.604	0.625	0.0210
Pneumonia	0.706	0.703	-0.0030
Pneumothorax	0.788	0.798	0.0100
Consolidation	0.754	0.781	0.0270
Edema	0.872	0.892	0.0200
Emphysema	0.712	0.743	0.0310
Fibrosis	0.659	0.71	0.0510
Pleural Thickening	0.726	0.742	0.0160
Hernia	0.728	0.761	0.0330

Table 1: Per-Disease AUC Between Simple and Complex ChestMNIST Models

C. RetinaMNIST Models Results and Performance

Both the parameter-heavy (3 convolutional layers with large dense layers) and parameter-efficient (4 convolutional layers with Global Average Pooling) architectures achieved identical test accuracy of 53.00% and nearly identical test AUC (81.55%), demonstrating that dramatic parameter reduction did not compromise overall performance on the 5-class diabetic retinopathy severity grading task. However, both models exhibited severe class imbalance issues, completely failing to detect Grade 4 (proliferative DR) cases.

The parameter-heavy model (Figure 6) trained for 38 epochs and exhibited moderate overfitting, with a 6.57 percentage point accuracy gap and 5.67 percentage point AUC gap between training and validation. The parameter-efficient model (Figure 7) required 46 epochs and remarkably showed negative overfitting gaps (validation outperformed training), suggesting the Global Average Pooling architecture provided superior regularization without additional dropout or data augmentation.

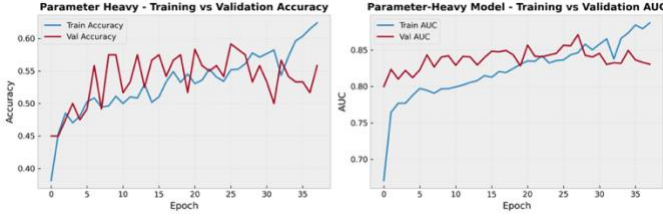


Figure 6: Parameter-Heavy RetinaMNIST Model Learning Curves

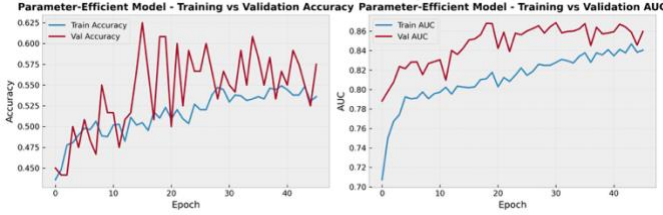


Figure 7: Parameter-Efficient Model Learning Curves

Per-class analysis reveals critical deployment concerns. The simple model achieved decent performance on Grade 0 (78.2%) but minimal accuracy on Grade 1 (6.5%), while the parameter-efficient model improved Grade 0 detection (81.0%) and Grade 1 performance (13.0%) at the cost of worse Grade 2 accuracy (31.5% vs. 46.7%). Most critically, both architectures completely failed to detect Grade 4 (0% accuracy), the most urgent severity level requiring immediate clinical intervention. The parameter-efficient model demonstrated better discrimination for extreme grades (0 and 3), achieving 81.0% and 57.4% accuracy, respectively, compared to the simple model's 78.2% and 44.1%.

Accuracy Comparison: Simple vs Parameter Efficient Model
Diabetic Retinopathy Severity Grading

Severity Grade	Simple Model	Parameter Efficient	Improvement
Grade 0	0.782	0.81	0.0280
Grade 1	0.065	0.13	0.0650
Grade 2	0.467	0.315	-0.1520
Grade 3	0.441	0.574	0.1330
Grade 4	0.0	0.0	0.0000

Table 2: Per-class accuracy comparison between parameter-heavy and parameter-efficient models for diabetic retinopathy severity grading.

V. RESULTS

A. Model Evaluation Strategy

Both datasets revealed fundamental limitations of training CNNs from scratch on small medical imaging datasets. ChestMNIST demonstrated acceptable performance (94.77% accuracy, 82-84% AUC) but required 78,468 training images. RetinaMNIST's 1,080 training images proved insufficient, achieving only 53% accuracy with complete Grade 4 detection failure, a critical clinical shortcoming.

Architectural complexity showed minimal impact. ChestMNIST's deeper architecture improved AUC by 1.28% but maintained identical accuracy. RetinaMNIST's 67x parameter reduction (16.9M→0.25M parameters) maintained equivalent performance, suggesting the dataset size, not model capacity, was the limiting factor.

Class imbalance mitigation failed on RetinaMNIST. Focal loss, class weights, and batch normalization either collapsed to majority-class prediction or reduced performance below baseline, indicating standard techniques are insufficient for datasets with small amounts of samples per minority class.

CONCLUSION

Small medical imaging datasets without transfer learning face critical limitations for clinical deployment. While ChestMNIST's 78,468 images (training set) enabled robust multi-label pathology detection, RetinaMNIST's 1,080 (training) images produced models that completely failed to detect Grade 4 diabetic retinopathy, the most urgent severity requiring immediate intervention. This failure highlights that rare but critical conditions demand either substantially larger datasets or transfer learning from pre-trained models like ImageNet or domain-specific medical networks.

Standard class imbalance techniques proved ineffective, suggesting small medical datasets require fundamentally different approaches: transfer learning, federated learning across institutions, or synthetic data generation. For real-world deployment, practitioners should prioritize transfer learning over training from scratch when dataset size is limited, particularly for tasks where minority class detection carries clinical significance.

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